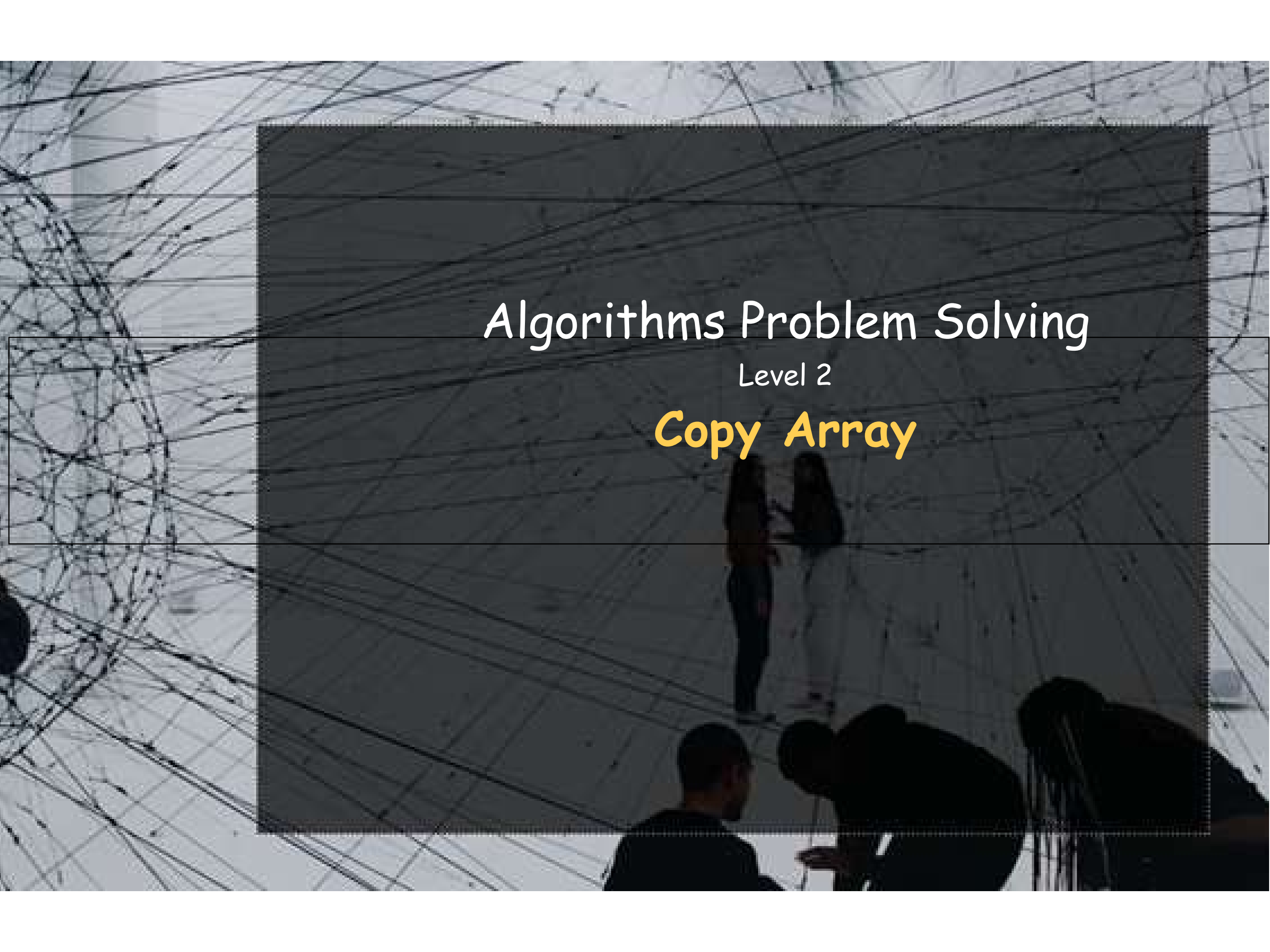


Algorithms Problem Solving

Level 2

Copy Array



Algorithms Problem Solving

Level 2

Copy Array

Definition

- ◆ This program fills an array with random numbers from `1` to `100`.
- ◆ It then copies all elements from the first array into a second array.
- ◆ Finally, it prints both arrays to verify that the copy was successful.

Explanation

- ◆ ``FillArrayWithRandomNumber()`` fills ``arr1`` with random numbers.
- ◆ ``CoopyArray()`` loops through all elements of ``arr1`` and assigns each element to the corresponding position in ``arr2``.
- ◆ ``PrintArrayElements()`` prints the elements of an array.
- ◆ The same ``arrayLength`` is used for both arrays because they contain the same number of elements.
- ◆ The copy is performed element by element:
`arr2[i] = arr1[i];`
- ◆ This means the value at index ``i`` in ``arr1`` is copied to index ``i`` in ``arr2``.

Steps in main()

1. Ask the user for the number of elements.
2. Fill `arr1` with random numbers from `1` to `100`.
3. Print the elements of `arr1`.
4. Loop through all elements of `arr1`.
5. Copy each element into the corresponding position in `arr2`.
6. Print the elements of `arr2`.

💡 Example

◆ Number of elements: `10`

◆ `Array 1`:

42 68 35 1 70 25 79 59 63 65

◆ Each element is copied:

arr2[0] = arr1[0] → 42

arr2[1] = arr1[1] → 68

arr2[2] = arr1[2] → 35

◆ After copying:

Array 2:

42 68 35 1 70 25 79 59 63 65

➡ Result

Array 1 Elements: 42 68 35 1 70 25 79 59 63 65

Array 2 Elements: 42 68 35 1 70 25 79 59 63 65

✈ Note

- ◆ `CoopyArray()` receives both arrays:

```
void CoopyArray(int arr1[], int arr2[], int& arrLength)
```
- ◆ `arr1` is the source array.
- ◆ `arr2` is the destination array.
- ◆ `arrLength` tells the function how many elements should be copied.
- ◆ The function does not create a new array. It copies the values into an existing array.
- ◆ The function name has a spelling mistake: `CoopyArray` should ideally be `CopyArray`.

⊘ Warning

- ◆ ``arr2`` must have enough space to hold all copied elements.
- ◆ Since both arrays have a maximum size of ``100``, ``arrayLength`` should not exceed ``100``.
- ◆ ``ReadNumber()`` and ``ReadArrayElements()`` are defined but not used.
- ◆ ``cmath`` and ``corecrt_math_defines.h`` are unnecessary for this program.
- ◆ ``rand()`` is not seeded with ``srand()``, so the same pseudo-random sequence may appear between program runs.

Comparison

`arr1`

- ✓ Source array.
- ✓ Contains the original random numbers.
- ✓ Example: `42 68 35 1`.

`arr2`

- ✓ Destination array.
- ✓ Receives copied values from `arr1`.
- ✓ Example after copying: `42 68 35 1`.

✦ Summary

- ✓ `arr1` contains the original elements.
- ✓ `arr2` receives a copy of those elements.
- ✓ Use a loop to copy array elements one by one.
- ✓ `arr2[i] = arr1[i]` copies the value at index `i`.
- ✓ Both arrays contain the same values after the copy.

Thank you 😊